

Lecture 1: Coordinates, Lines, and Circles

Euclidean geometry studies shapes through size, angle, and relationship; analytic geometry studies the same shapes by assigning numbers to points, so that geometric facts turn into algebraic equations. A point becomes an ordered pair (x, y) , a curve becomes an equation in x and y , and questions about length, parallelism, and intersection become questions we can settle by algebra. The coordinate plane does not replace Euclidean geometry; it gives a second language for the same ideas. This lecture builds the dictionary between the two.

1. A Dictionary Between Two Languages

Euclidean geometry	Coordinate geometry
point	ordered pair (a, b)
curve / shape (set of points satisfying a property)	equation in x, y (set of solutions)
distance between two points	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
midpoint of a segment	$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
circle (all points at fixed distance from a center)	equation $(x - h)^2 + (y - k)^2 = r^2$
steepness / direction of a line	slope $m = \frac{y_2 - y_1}{x_2 - x_1}$
two points determine a line	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
two lines intersect in one point	the system $y = m_1x + b_1$, $y = m_2x + b_2$ has the unique solution $x = \frac{b_2 - b_1}{m_1 - m_2}$ when $m_1 \neq m_2$
parallel lines	equal slopes $m_1 = m_2$
perpendicular lines	$m_1 m_2 = -1$
median from a vertex	segment from (x_1, y_1) to $\left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right)$
centroid	average of the vertices: $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$
altitude from a vertex	line through that vertex with slope $-1/m$, where m is the slope of the opposite side
orthocenter	common solution of the three altitude equations

This dictionary is the foundation of analytic geometry, and we will return to it all term: a geometric statement such as “the lines are perpendicular” becomes an algebraic statement, “the slopes multiply to -1 .” The last four entries are developed in the final two sections.

2. Points as Coordinates

A point (a, b) is determined by two signed distances: a is the signed distance from the y -axis, and b is the signed distance from the x -axis. The two axes cut the plane into four **quadrants**, numbered I–IV counterclockwise starting from the upper right, where both coordinates are positive.

Example. The point $C(3, 2)$ means “move 3 units right and 2 units up.”

Translation rules. Shifting a point is addition in one coordinate:

- vertical shift: $(a, b) \mapsto (a, b + c)$,
- horizontal shift: $(a, b) \mapsto (a + c, b)$.

Exercise 1. Plot $(3, -2)$, $(-4, 1)$, and $(0, 5)$, and say which quadrant (or axis) each lies in. Where does each point go under $(a, b) \mapsto (a + 3, b - 1)$?

3. Distance and Midpoint

Definition. The distance between $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

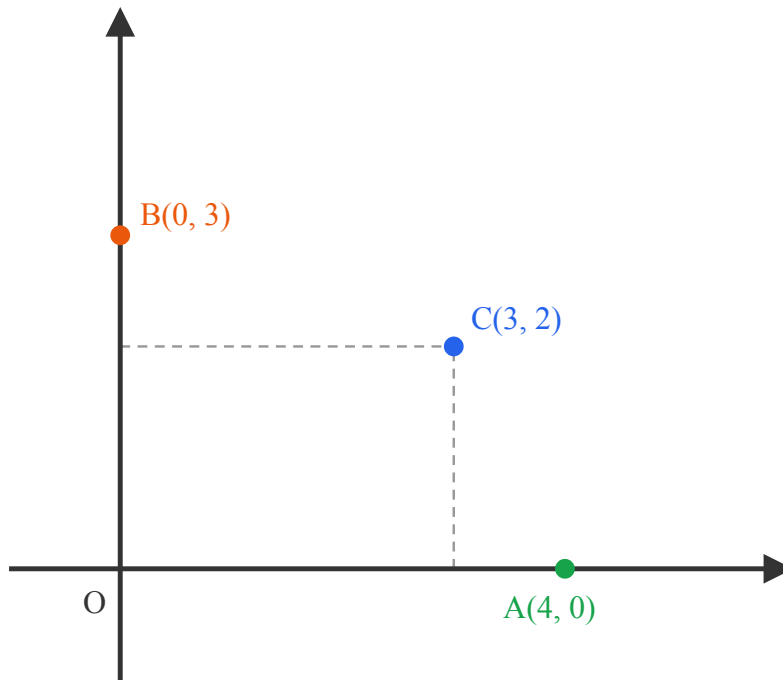


Figure 1: Coordinate plane showing a general point $C(3,2)$, a point $A(4,0)$ on the x-axis, and a point $B(0,3)$ on the y-axis

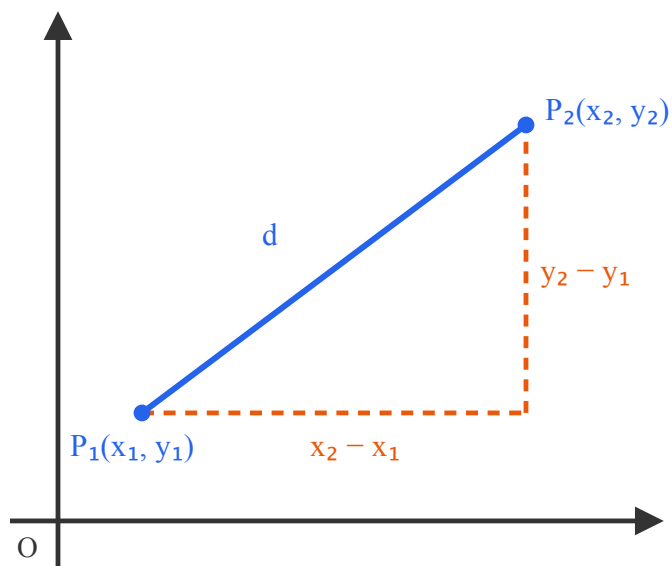


Figure 2: Right triangle with legs $x_2 - x_1$ and $y_2 - y_1$, hypotenuse the segment from P_1 to P_2 , used to derive the distance formula

Proof. The horizontal and vertical legs of the right triangle have lengths $|x_2 - x_1|$ and $|y_2 - y_1|$, respectively. By the Pythagorean theorem,

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

Taking square roots gives the stated formula. ■

Midpoint formula. The midpoint of (x_1, y_1) and (x_2, y_2) is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right),$$

the average of the corresponding coordinates. Distance uses the *differences* of coordinates; the midpoint uses their *averages*.

Exercise 2. Find the distance between $(1, 1)$ and $(4, 5)$.

Exercise 3. Show that $(0, 0)$, $(3, 0)$, and $(0, 4)$ form a right triangle using the distance formula.

Exercise 4. Find the midpoint of $(-3, 4)$ and $(5, -2)$, and verify by the distance formula that it is equidistant from both.

Exercise 5. A point $(x, 0)$ on the x -axis is equidistant from $(1, 2)$ and $(5, -4)$. Find x .

4. Circles

Definition (Euclidean). A circle with center C and radius r is the set of all points whose distance from C is exactly r .

Coordinate formula. A point (x, y) lies on the circle with center (h, k) and radius r if and only if its distance from (h, k) is r ; equivalently,

$$(x - h)^2 + (y - k)^2 = r^2.$$

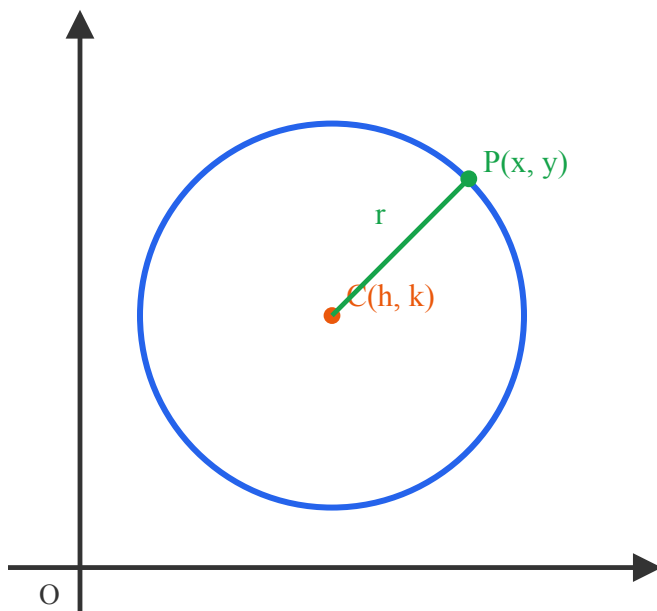


Figure 3: Circle with center $C(h, k)$, radius r , and a point $P(x, y)$ on the circle

Example. The circle centered at the origin with radius 1 has equation $x^2 + y^2 = 1$, the unit circle.

Exercise 6. Write the equation of the circle with center $(2, -1)$ and radius 3.

Exercise 7. Does the point $(4, 3)$ lie on, inside, or outside the circle $x^2 + y^2 = 20$?

Exercise 8. The equation $x^2 + y^2 - 4x + 6y - 12 = 0$ represents a circle. Complete the square to write it in standard form $(x - h)^2 + (y - k)^2 = r^2$, then find its center and radius.

Exercise 9. Find the equation of the circle passing through the three points $(0, 0)$, $(4, 0)$, and $(0, -2)$.

Hint. Two routes, one in each language. Algebraically: substitute the three points into $x^2 + y^2 + Dx + Ey + F = 0$ and solve for D , E , F . Geometrically: the angle at $(0, 0)$ is a right angle, so the segment joining the other two points is a diameter — now use the midpoint formula.

5. Lines and Slope

Definition. For two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$, the slope of the line through them is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

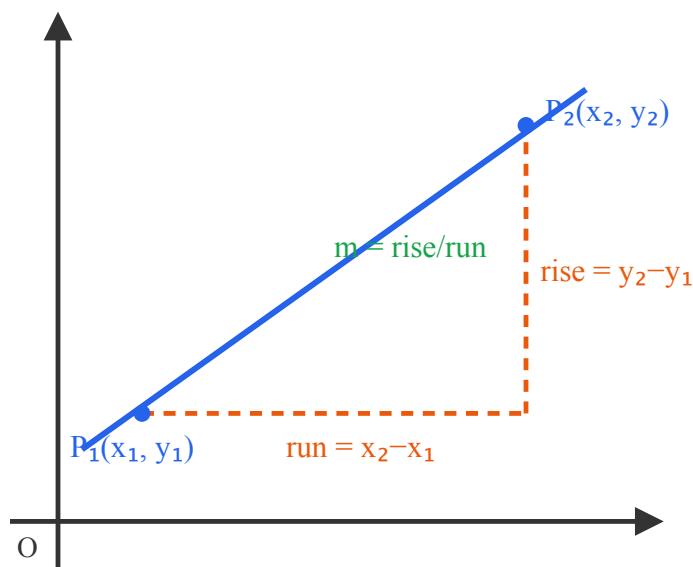


Figure 4: General two points with rise and run forming the slope of a line

Sign of the slope.

- $m > 0$: the line rises from left to right.
- $m < 0$: the line falls from left to right.
- $m = 0$: the line is horizontal.
- undefined: the line is vertical.

Examples. A horizontal line $y = b$ has no rise, so $m = 0$. A vertical line $x = a$ has no run: any two of its points have $x_1 = x_2$, the denominator in the definition is zero, and the slope is undefined. Vertical lines will keep turning up as the exception to statements about slope.

Proposition. Any two points on the same line give the same slope; thus the slope of a line is well defined.

Proof. Let A , B , C lie on the line, with B and C on the same side of A . Drop a vertical segment from B to the horizontal line through A , meeting it at B' , and likewise from C , meeting it at C' . Then $\triangle AB'B$ and

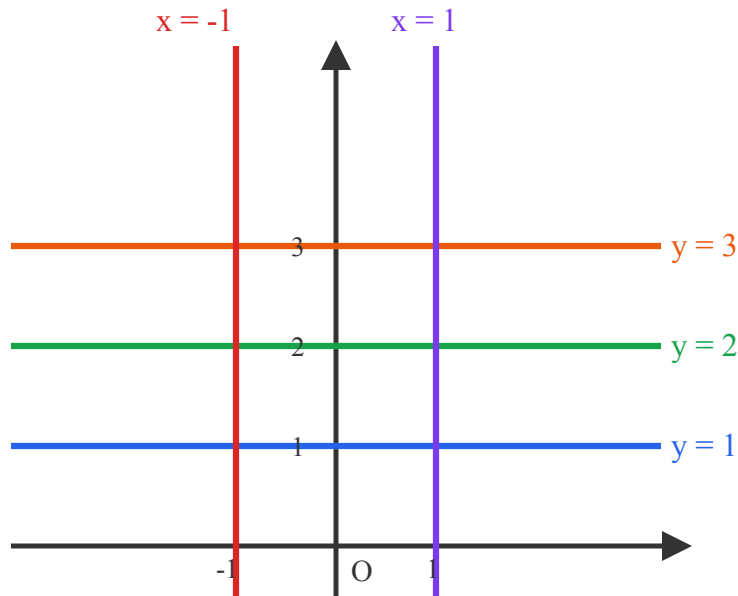


Figure 5: Horizontal lines $y=1, y=2, y=3$ and vertical lines $x=1, x=-1$ on a coordinate plane

$\triangle AC'C$ are right triangles sharing the angle at A , so they are similar. Similar triangles have proportional side lengths, hence

$$\frac{BB'}{AB'} = \frac{CC'}{AC'},$$

which says the ratio $\frac{|\text{rise}|}{|\text{run}|}$ is the same for both pairs. The signs agree as well: travelling from A toward B or toward C we move the same way horizontally and the same way vertically, so rise and run carry the same signs in both ratios. Therefore the slope does not depend on which two points of the line we choose. ■

Exercise 10. Compute the slope between $(-1, 3)$ and $(2, -3)$. Is the line rising or falling?

Exercise 11. The points $(1, 2)$, $(3, k)$, and $(7, 14)$ are collinear. Find k .

6. Equations of a Line

Point-slope formula. The line through (x_1, y_1) with slope m is

$$y - y_1 = m(x - x_1).$$

Rearranging gives the familiar slope-intercept form

$$y = mx + \underbrace{(y_1 - mx_1)}_b.$$

Two-point formula. If a line passes through (x_1, y_1) and (x_2, y_2) , then its slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

and substituting into the point-slope formula yields

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1).$$

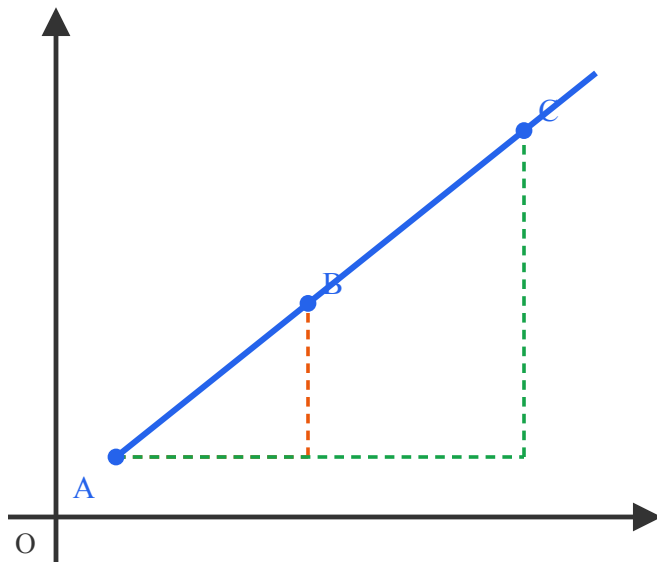


Figure 6: Similar right triangles from three collinear points showing slope is well defined

General form. Every line — vertical ones included — can be written as

$$Ax + By + C = 0, \quad A, B \text{ not both } 0.$$

If $B \neq 0$ this rearranges to $y = mx + b$ with $m = -A/B$; if $B = 0$ it is the vertical line $x = -C/A$. The general form is worth knowing precisely because it does not break down in the vertical case.

Example. The line through $(1, 2)$ and $(3, 6)$ has slope

$$m = \frac{6 - 2}{3 - 1} = 2,$$

so

$$y - 2 = 2(x - 1),$$

which simplifies to $y = 2x$.

Exercise 12. Find the equation of the line through $(0, 5)$ and $(2, 1)$.

Exercise 13. A line through $(a, 2)$ and $(3, b)$ has slope 2 and passes through $(0, 1)$. Find a and b .

7. Parallel and Perpendicular Lines

Proposition. Two nonparallel, non-vertical lines intersect in exactly one point.

Proof. Solving

$$m_1x + b_1 = m_2x + b_2$$

for x gives

$$x = \frac{b_2 - b_1}{m_1 - m_2},$$

a unique solution whenever $m_1 \neq m_2$. If $m_1 = m_2$ the lines are parallel (or coincident), and they do not meet in exactly one point. ■

(If one of the lines is vertical, say $x = a$, the argument is even shorter: substitute $x = a$ into the other equation.)

Exercise 14. Find the intersection of $y = 2x + 1$ and $y = -x + 4$.

Parallel lines. Two lines

$$\ell_1 : y = m_1x + b_1, \quad \ell_2 : y = m_2x + b_2$$

are parallel if and only if $m_1 = m_2$.

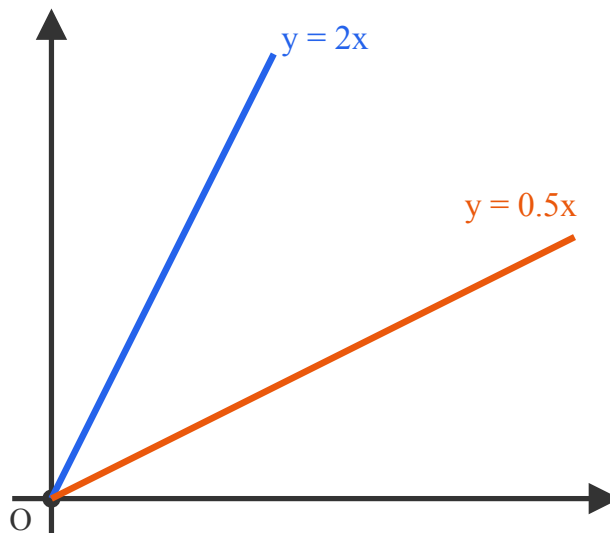


Figure 7: Two lines through the origin with different slopes meeting only at the origin

Perpendicular lines. Two lines with slopes m_1 and m_2 are perpendicular if and only if

$$m_1m_2 = -1.$$

Proof. Place the common point of the two lines at the origin O and label them so that $m_1 > 0$. Neither line is horizontal, since the perpendicular to a horizontal line is vertical and has no slope; so $m_2 < 0$. Moving one unit to the right along each line gives

$$A = (1, m_1) \in \ell_1, \quad B = (1, m_2) \in \ell_2,$$

and we set $F = (1, 0)$. Then $OF = 1$, $AF = m_1$, and $FB = -m_2$, all positive lengths.

The segment AB is vertical and OF is horizontal, so both $\triangle OFA$ and $\triangle BFO$ have a right angle at F . The ray OF splits the right angle $\angle AOB$, so $\angle FOA$ and $\angle FOB$ are complementary; and in the right triangle BFO , the angles $\angle FBO$ and $\angle FOB$ are complementary too. Hence $\angle FOA = \angle FBO$, and the two triangles are similar. Therefore

$$\frac{OF}{FB} = \frac{AF}{OF}, \quad \text{so} \quad OF^2 = AF \cdot FB.$$

Substituting the lengths above gives $1 = m_1(-m_2)$, that is, $m_1m_2 = -1$.

Conversely, suppose $m_1m_2 = -1$ with $m_1 > 0 > m_2$. Then $OF^2 = AF \cdot FB$, so the right triangles OFA and BFO have proportional legs about their right angles and are therefore similar. This gives $\angle FOA = \angle FBO$, which is complementary to $\angle FOB$, so $\angle AOB = \angle FOA + \angle FOB = 90^\circ$ and the lines are perpendicular. ■

Example. The lines

$$y = \frac{1}{2}x + 1 \quad \text{and} \quad y = -2x + 3$$

have slopes $\frac{1}{2}$ and -2 , and

$$\frac{1}{2} \cdot (-2) = -1.$$

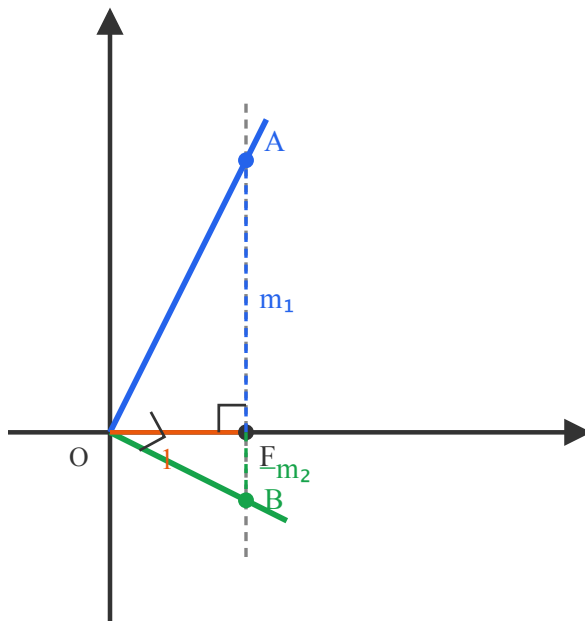


Figure 8: Two perpendicular lines through O read off at the same run, with similar triangles OFA and BFO

Therefore the lines are perpendicular.

Remark (a form with no fractions). Clearing denominators in $m_1 m_2 = -1$ gives a version of the perpendicularity test that never divides by zero. The line through $P(x_P, y_P)$ and $Q(x_Q, y_Q)$ is perpendicular to the line through $R(x_R, y_R)$ and $S(x_S, y_S)$ if and only if

$$(x_Q - x_P)(x_S - x_R) + (y_Q - y_P)(y_S - y_R) = 0.$$

Indeed, $\frac{y_Q - y_P}{x_Q - x_P} \cdot \frac{y_S - y_R}{x_S - x_R} = -1$ becomes this equation after multiplying through by both denominators. Unlike the slope form, it stays correct when one of the lines is vertical, which is exactly the case that will come up with altitudes in Section 9.

Exercise 15. Find the line through $(1, 1)$ perpendicular to $y = 3x - 2$.

Exercise 16. Find the equation of the line through the intersection of $y = 2x + 1$ and $y = -x + 4$ that is perpendicular to $y = \frac{1}{2}x - 3$.

8. The Centroid of a Triangle

Definition (Euclidean). A median is a segment joining a vertex of a triangle to the midpoint of the opposite side. The centroid G is the point where the three medians intersect.

Proposition. If a triangle has vertices (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , then its centroid is

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).$$

Moreover, G divides each median in the ratio $2 : 1$, measured from the vertex.

Proof. Let $A = (x_1, y_1)$, and let M_1 be the midpoint of the opposite side, so

$$M_1 = \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2} \right).$$

The point two-thirds of the way from A to M_1 is

$$A + \frac{2}{3}(M_1 - A) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).$$

This expression is symmetric in the three vertices, so the same point lies $2/3$ of the way along every median. Hence all three medians pass through it, and the point is the centroid. ■

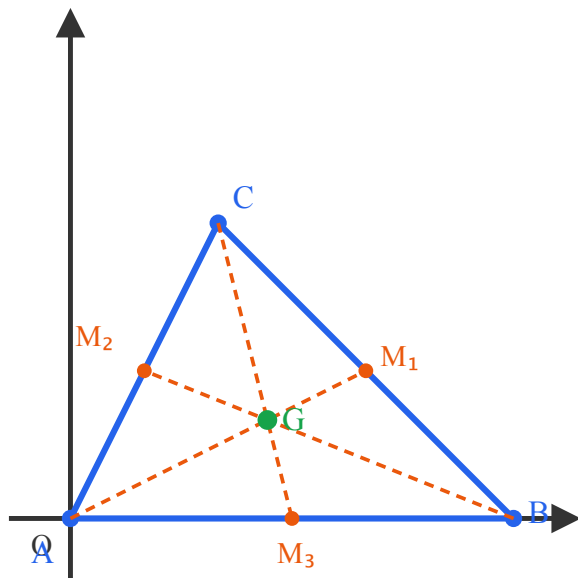


Figure 9: Triangle ABC with its three medians meeting at the centroid G

Example. For the triangle with vertices $(0,0)$, $(6,0)$, and $(3,6)$, the centroid is

$$\left(\frac{0+6+3}{3}, \frac{0+0+6}{3} \right) = (3,2).$$

Exercise 17. Find the centroid of the triangle with vertices $(1,1)$, $(5,1)$, and $(3,7)$.

Exercise 18. Two vertices of a triangle are $(0,0)$ and $(4,2)$, and the centroid is $(2,3)$. Find the third vertex.

Exercise 19 (Challenge). Prove that the centroid divides each median in the ratio $2 : 1$ by a purely geometric argument, without coordinates.

Hint. Apply the midsegment theorem to the segment joining the midpoints of two sides.

Exercise 20 (Challenge). For a general triangle with vertices (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , show directly that all three medians pass through

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right),$$

without assuming the $2 : 1$ ratio result.

Hint. Three points P, Q, R are collinear if and only if

$$\frac{y_Q - y_P}{x_Q - x_P} = \frac{y_R - y_P}{x_R - x_P}.$$

Apply this to the vertex (x_1, y_1) , the midpoint $M_1 = \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2} \right)$, and the centroid; then repeat for the other two medians by symmetry.

9. Altitudes and the Orthocenter

The centroid is one of the classical centers of a triangle. Another is the orthocenter, where the three altitudes meet.

Definition. An **altitude** of a triangle is the line through a vertex perpendicular to the opposite side. The **orthocenter** H is the point where the three altitudes meet.

Because an altitude is defined by a perpendicularity condition, altitude problems are exactly where the results of Section 7 pay off — including the fraction-free form of the perpendicularity test, since an altitude drawn to a horizontal side is vertical and has no slope.

Exercise 21. Let $A = (0, 0)$, $B = (6, 0)$, and $C = (2, 8)$. Find the equations of the three altitudes of triangle ABC , and show that they meet at a single point. What is that point?

Hint. The side AB is horizontal, so the altitude from C is vertical: write it as $x = \text{constant}$ rather than trying to use a slope.

Exercise 22 (Challenge). Let $A = (x_1, y_1)$, $B = (x_2, y_2)$, $C = (x_3, y_3)$.

- (a) Using the remark at the end of Section 7, show that (x, y) lies on the altitude from A if and only if

$$(x - x_1)(x_3 - x_2) + (y - y_1)(y_3 - y_2) = 0,$$

and write the two analogous equations for the altitudes from B and from C , keeping the same cyclic pattern $A \rightarrow B \rightarrow C \rightarrow A$ of the indices.

- (b) Expand all three equations and add them. Show that the sum is identically zero — every term cancels, for every point (x, y) .
- (c) Conclude that the three altitudes are concurrent: any point lying on two of them automatically lies on the third. That point is the orthocenter H .

Remark. Part (b) proves the concurrency of the altitudes without ever solving for H — a good example of algebra doing geometric work. Finding a closed formula for H is possible but much messier than proving that it exists.